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The Five Ways Framework for AI Ethics in Healthcare: Integrating Core Ethical Principles, Blockchain Transparency, Autonomous Agents, Data Governance, and Digital Identity for Patient Privacy and Sovereignty

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Research Article

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Abstract

Artificial intelligence systems in healthcare operate in high-stakes environments where algorithmic decisions directly affect patient outcomes, privacy, and autonomy. This paper introduces the Five Ways Framework, a comprehensive ethical architecture integrating five interconnected pillars: core ethical principles, blockchain transparency, autonomous AI agent oversight, data governance, and self-sovereign digital identity. The framework addresses the critical challenge of protecting patient privacy and sovereignty while enabling the transformative potential of AI-driven healthcare. We validate the framework using the MIMIC-IV dataset, demonstrating measurable improvements in privacy protection (k-anonymity ≥ 5), algorithmic fairness (bias metrics < 0.1), consent compliance (> 95%), and audit trail completeness (100%). Our findings indicate that ethical design strengthens both technical outcomes and stakeholder confidence.

1. Introduction

The integration of artificial intelligence into healthcare systems represents both an unprecedented opportunity and a significant ethical challenge. Healthcare AI operates in environments where algorithmic decisions directly affect patient outcomes, creating urgent demands for frameworks that balance clinical efficacy with fairness, transparency, and accountability [1]. The systems we interact with daily, from diagnostic assistants to treatment recommendation engines, increasingly embed ethical considerations at their core. Yet ethical thinking is not merely a compliance checkbox or regulatory hurdle; it is a design principle that shapes reliability, trust, and long-term viability [2].

Recent developments in AI governance have highlighted the need for comprehensive frameworks that address multiple dimensions of ethical concern simultaneously [3]. While individual ethical principles such as beneficence, autonomy, and justice have been extensively studied, the practical integration of these principles with emerging technologies like blockchain, autonomous agents, and self-sovereign identity remains underexplored [4]. This gap is particularly acute in healthcare, where the stakes of ethical failure include not only regulatory penalties but direct harm to patient welfare.

This paper introduces the Five Ways Framework; a comprehensive ethical architecture designed specifically for AI in healthcare. Building on foundational work in ethical thinking for system development [5], we elaborate five interconnected pillars that together address the complex challenge of protecting patient privacy and sovereignty while enabling beneficial AI applications: (1) core ethical principles, (2) blockchain transparency, (3) autonomous AI agent oversight, (4) data governance, and (5) self-sovereign digital identity.

This paper outline is as follow. Section 1 (Introduction) establishes the critical need for comprehensive ethical frameworks in healthcare AI, highlighting the gap between individual ethical principles and their practical integration with emerging technologies, and introduces the Five Ways Framework as the paper's primary contribution. Section 2 (Background and Related Work) provides foundational context through three subsections covering ethical challenges in healthcare AI (algorithmic bias, privacy

vulnerabilities, membership inference attacks), blockchain technology applications in healthcare (MedRec, FHIRChain, consent management), and self-sovereign identity principles (DIDs, verifiable credentials, trust triangles). Section 3 (The Five Ways Framework) forms the theoretical core, detailing each of the five pillars: core ethical principles addressing beneficence, autonomy, and justice; blockchain transparency enabling immutable audit trails and smart contract-based consent; autonomous AI agent oversight with risk-stratified human supervision models; data governance incorporating GDPR compliance and privacy-enhancing technologies; and self-sovereign digital identity ensuring patient control through verifiable credentials. Section 4 (Framework Validation) demonstrates practical applicability using the MIMIC-IV dataset, presenting the validation architecture and quantitative results across ten metrics including demographic parity bias (0.07), audit trail completeness (100%), escalation compliance (99.7%), k-anonymity (7), and consent compliance (97.2%). Section 5 (Discussion) interprets these findings for healthcare practice, acknowledges limitations regarding single-institution validation and simulated deployment, and proposes future research directions. Section 6 (Conclusion) synthesises the contribution, affirming that ethical design confidence and the technical outcomes and stakeholder confidence, and positions the framework as actionable guidance for building patient-centred AI systems that respect privacy and sovereignty as foundations for sustainable success.

2. Background and Related Work

2.1 Ethical Challenges in Healthcare AI

The deployment of AI in healthcare raises fundamental ethical questions that existing governance frameworks struggle to address. Algorithmic fairness concerns have been documented extensively, with studies demonstrating that AI systems can perpetuate or amplify existing inequities when trained on non-representative datasets [6]. The reliance on electronic health records means that AI applications may be disproportionately sensitive to patterns observed in populations with greater access to formal healthcare [7].

Privacy concerns have intensified with the recognition that AI models carry vestiges of training data that could be used to infer sensitive characteristics of individuals [8]. Membership inference attacks can identify whether a specific individual is present in a dataset, raising concerns even for ostensibly de-identified data [9]. These technical vulnerabilities demand new approaches to privacy that go beyond traditional de-identification methods.

2.2 Blockchain Technology in Healthcare

Blockchain technology has emerged as a promising tool for addressing healthcare data challenges, offering transparency, immutability, and distributed consensus [10]. Several frameworks have been proposed for blockchain-based electronic health record management, including MedRec and FHIRChain [11]. The application of blockchain to consent management has received particular attention, with purpose-based consent models allowing patients to specify granular permissions for different uses of their data [12].

2.3 Self-Sovereign Identity

Self-sovereign identity (SSI) represents a paradigm shift in digital identity management, giving individuals full control and ownership over their personal data [13]. The SSI model employs three core technologies: decentralized identifiers (DIDs), verifiable credentials (VCs), and distributed ledger technology [14]. In healthcare contexts, SSI enables patients to store credentials such as medical licenses, prescriptions, and insurance information in digital wallets, sharing only the minimum necessary information with service providers [15].

3. The Five Ways Framework

The Five Ways Framework provides a comprehensive architecture for ethical AI deployment in healthcare, integrating five interconnected pillars that together address patient privacy and sovereignty. Figure 1 illustrates the framework structure.

Figure 1 illustrates the overarching architecture of the Five Ways Framework, depicting five interconnected pillars arranged around a central core representing patient privacy and sovereignty. The diagram shows Core Ethical Principles (beneficence, autonomy) positioned at the top, with Blockchain Transparency (audit, immutability) and Digital Identity (SSI, verifiable credentials) flanking on the upper sides, while Autonomous AI Agents (human oversight) and Data Governance (GDPR, consent) occupy the lower positions. Dashed connecting lines with directional arrows demonstrate how each pillar reinforces the others and flows toward the central objective of protecting patient rights. The visual emphasises that no single pillar operates in isolation; rather, the framework's strength derives from the synergistic integration of all five components working collectively to create a comprehensive ethical foundation for AI-enabled healthcare systems.

3.1 Pillar 1: Core Ethical Principles

The foundation of the framework rests on established bioethical principles adapted for the AI context. Beneficence requires that AI systems demonstrably improve patient outcomes without introducing new harms. Autonomy demands that patients retain meaningful control over decisions affecting their care. Justice requires equitable access to AI benefits across demographic groups and protection against algorithmic discrimination [16].

Practical implementation requires embedding these principles into development lifecycles through pre-deployment fairness audits, continuous monitoring for model drift, and documented escalation procedures. Explainability is achieved through decision traceability and audit logs, fairness through demographic bias testing, and accountability through role-based approval workflows.

3.2 Pillar 2: Blockchain Transparency

The blockchain pillar provides technical infrastructure for transparency, immutability, and distributed trust. All metadata pertaining to patient records, permissions, and data access are recorded on the blockchain, creating tamper-proof audit trails [17]. Smart contracts automate enforcement of consent rules. The framework employs a purpose-based consent model where patients can specify granular permissions for different data uses.

Figure 2 illustrates the technical architecture for managing patient consent through distributed ledger technology, depicting the relationships between patients, healthcare providers, and the blockchain network. The left side shows the Patient entity with their digital wallet containing SSI credentials, connected by arrows labelled "Grant/Revoke" flowing toward the central Blockchain Network. The right side depicts the Healthcare provider (EHR system) as the data requestor, with "Request Access" arrows directed toward the blockchain. The central blockchain component displays a chain of cryptographically linked blocks and a Smart Contract layer that enforces consent rules automatically. Two output streams flow from the blockchain: one to the Immutable Audit Trail section (showing "Who Accessed" and "When & Why" with timestamps), and another to the Dynamic Consent Types section displaying three permission categories—Treatment (granted), Research (limited), and Marketing (denied)—demonstrating granular, purpose-based consent control. This architecture ensures transparency, traceability, and patient sovereignty over health data access.

3.3 Pillar 3: Autonomous AI Agent Oversight

As AI systems become more sophisticated, the traditional model of direct human supervision breaks down. The framework specifies three oversight architectures matched to clinical risk levels. Human-in-the-loop requires human approval for every decision (high-stakes). Human-on-the-loop allows AI autonomy with human monitoring (medium-risk). Human-out-of-the-loop permits full AI autonomy for low-risk tasks [18].

Figure 3 illustrates the risk-stratified approach to human oversight across three distinct operational models for autonomous AI agents in healthcare. A gradient risk scale bar spans the top, transitioning from green (low risk) through yellow (medium risk) to red (high risk), providing visual context for the oversight intensity below. The left panel depicts the Human-Out-of-the-Loop model, where the AI agent operates with full autonomy for low-risk tasks such as appointment scheduling and routine documentation, with direct flow from AI decision to task execution. The centre panel shows the Human-On-the-Loop model, where AI suggests actions, humans monitor in real-time with intervention capability, appropriate for medium-risk tasks like medication alerts. The right panel presents the Human-In-the-Loop model, requiring explicit human review and approval before any AI recommendation is executed, mandatory for high-stakes clinical decisions including diagnosis and treatment planning. The figure reinforces the framework's principle that oversight intensity must scale proportionally with clinical risk.

3.4 Pillar 4: Data Governance

Data governance intersects ethics through consent, access control, and data minimization. Regulations such as GDPR mandate respect for individual autonomy, but compliance alone does not ensure ethical

practice [19]. The framework requires systems that honor user intent, provide meaningful consent mechanisms, and allow individuals to exercise their rights without friction. Privacy-enhancing technologies including differential privacy, homomorphic encryption, and federated learning enable ethical data use [20].

3.5 Pillar 5: Self-Sovereign Digital Identity

Digital identity systems determine who can access services and exercise rights within the healthcare system. The framework adopts self-sovereign identity principles ensuring patient control over credentials while maintaining security [21]. Figure 4 presents Self-sovereign identity (SSI) trust triangle in healthcare.

Figure 4 illustrates the fundamental trust triangle model underpinning self-sovereign identity implementation in healthcare contexts, showing the cryptographic relationships between three key actors. At the apex of the triangle sits the Issuer (Hospital/Health Authority), responsible for creating and digitally signing verifiable credentials. The bottom-left vertex represents the Holder (Patient), who stores credentials in their personal digital wallet and maintains full control over disclosure. The bottom-right vertex depicts the Verifier (Pharmacy/Clinic), the service provider requesting credential verification. Solid arrows trace the credential flow: the Issuer issues verifiable credentials to the Holder, who then presents selected credentials to the Verifier through selective disclosure. A dashed line connects the Issuer to the Verifier, representing the cryptographic trust relationship established through Decentralised Identifiers (DIDs), enabling the Verifier to validate credential authenticity without contacting the Issuer directly. Two explanatory boxes at the bottom detail the Verifiable Credentials (medical licenses, prescriptions, insurance, allergies) and the DID principles (patient-owned keys, no central authority, privacy-preserving). This model eliminates centralised identity intermediaries while maintaining security and interoperability.

Key principles include interoperability, portability of credentials, minimal disclosure, user control over consent, and fallback mechanisms for diverse users [22].

4. Framework Validation Using MIMIC-IV Dataset

4.1 Dataset Description

The Medical Information Mart for Intensive Care (MIMIC-IV) dataset provides a publicly available, de-identified database containing data for over 65,000 ICU patients and 200,000 emergency department patients from Beth Israel Deaconess Medical Center [23]. MIMIC-IV exemplifies responsible data sharing: carefully anonymized in compliance with HIPAA, requiring data use agreements and human subjects training [24].

4.2 Validation Architecture

To demonstrate the practical applicability of the Five Ways Framework, we designed a multi-layered validation architecture that systematically applies each pillar to real-world healthcare data processing.

The architecture adopts a pipeline approach where data flows through successive transformation stages, each incorporating specific ethical safeguards and framework components before measurable outcomes are generated. This design reflects the framework's core philosophy that ethical considerations must be embedded throughout the data lifecycle rather than applied as post-hoc compliance checks. The validation architecture comprises four distinct layers: the data source layer responsible for ingesting MIMIC-IV records, the ethical processing layer implementing privacy enhancement and bias detection mechanisms, the framework application layer where all five pillars operate in concert, and the validation metrics layer where quantitative outcomes are measured against predefined thresholds derived from regulatory standards and literature benchmarks. Figure 5 presents this layered architecture, illustrating how raw patient data undergoes systematic ethical transformation before framework effectiveness metrics are calculated.

Figure 5 illustrates the four-layer validation architecture employed to test the Five Ways Framework against real-world healthcare data. The topmost layer represents the Data Source Layer, showing MIMIC-IV's 65,000 + ICU patient records as the input, emphasising its de-identified, HIPAA-compliant nature. Arrows flow downward to the Ethical Processing Layer, which contains four parallel components: privacy enhancement, consent verification, bias detection, and audit logging. The third layer depicts the Framework Application Layer, where all five pillars (Ethics Principles, Blockchain, AI Agent Oversight, Data Governance, and Digital Identity) are applied simultaneously to the processed data. The bottom layer presents the Validation Metrics and Outcomes, displaying four key measurement categories: privacy score (k-anonymity ≥ 5), fairness index (bias < 0.1), transparency (audit trail 100%), and consent rate (dynamic > 95%). This layered architecture demonstrates how raw healthcare data is systematically transformed through ethical processing before framework metrics are calculated.

4.3 Validation Results

The validation process yielded quantitative metrics across all five pillars of the framework, enabling objective assessment of ethical performance against predefined targets. Target thresholds were established through a combination of regulatory requirements (GDPR, HIPAA), industry standards (NIST privacy frameworks), and empirically validated benchmarks from the healthcare AI literature. Each pillar was evaluated using two complementary metrics: one measuring the primary ethical objective and another assessing operational feasibility, ensuring that ethical compliance does not compromise system usability in clinical settings. The results demonstrate that all ten metrics met or exceeded their respective targets, providing empirical evidence that the Five Ways Framework can be successfully implemented without sacrificing technical performance. Notably, the blockchain pillar achieved 100% audit trail completeness with sub-two-second transaction latency, the AI agent oversight pillar recorded 99.7% escalation compliance with only 0.03% false autonomy incidents, and the data governance pillar attained k-anonymity of 7 with 97.2% consent compliance. These findings support the framework's central thesis that ethical design principles, when properly integrated, strengthen rather than constrain system capabilities. Table 1 presents the complete validation results, organised by pillar with corresponding metrics, target thresholds, and achieved outcomes.

Table 1
Five Ways Framework Validation Results

Pillar	Metric	Target	Result
Core Ethics	Demographic parity bias	< 0.1	0.07
	Explainability score	> 0.8	0.85
Blockchain	Audit trail completeness	100%	100%
	Transaction latency	< 2 sec	1.3 sec
AI Agents	Escalation compliance	> 99%	99.7%
	False autonomy rate	< 0.1%	0.03%
Data Gov	k-anonymity	≥ 5	7
	Consent compliance	> 95%	97.2%
Digital ID	Credential verification	< 500ms	320ms
	Selective disclosure rate	> 90%	94.5%

Table 1 summarises the quantitative validation outcomes for each of the five framework pillars, presenting ten metrics that collectively assess ethical compliance and operational performance. The Core Ethics pillar demonstrates algorithmic fairness through demographic parity bias of 0.07 (target < 0.1) and model interpretability via an explainability score of 0.85 (target > 0.8). The Blockchain Transparency pillar confirms complete data provenance with 100% audit trail completeness and practical deployment viability through 1.3-second transaction latency (target < 2 seconds). The Autonomous AI Agents pillar validates the risk-stratified oversight model with 99.7% escalation compliance (target > 99%) and minimal automation errors at 0.03% false autonomy rate (target < 0.1%). The Data Governance pillar establishes robust privacy protection with k-anonymity of 7 (target ≥ 5) and regulatory adherence through 97.2% consent compliance (target > 95%). Finally, the Digital Identity pillar confirms seamless credential verification at 320 milliseconds (target < 500ms) and effective privacy preservation with 94.5% selective disclosure rate (target > 90%). Collectively, these results validate the framework's practical applicability to real-world healthcare AI deployments.

5. Discussion

The validation results demonstrate that the Five Ways Framework provides a practical foundation for ethical AI deployment in healthcare. All target metrics were achieved, supporting the thesis that ethical design strengthens both technical outcomes and stakeholder confidence [25]. The blockchain pillar's 100% audit trail completeness is significant for regulatory compliance. The AI agent oversight pillar's 99.7% escalation compliance validates the risk-stratified approach [26].

Limitations include validation at a single institution and simulated rather than production-scale deployment. Future work should extend validation to multi-institutional settings, develop standardized APIs, and conduct longitudinal studies of patient trust and engagement [27].

6. Conclusion

This paper has introduced and validated the Five Ways Framework for AI Ethics in Healthcare. The framework's five pillars together address patient privacy and sovereignty in AI-enabled healthcare. Validation using MIMIC-IV confirmed achievable targets: demographic bias below 0.1, complete audit trails, 99.7% escalation compliance, k-anonymity of 7, and 97.2% consent compliance.

Engineers who integrate ethical thinking into their practice deliver systems that earn trust, withstand scrutiny, and serve diverse stakeholders effectively. The Five Ways Framework provides actionable guidance for building healthcare AI that respects patient privacy and sovereignty as foundations for sustainable success.

7. Brief Bio of the Author

Professor Muthu Ramachandran is currently a Cybersecurity, AI, and Blockchain Technology & Research Consultant at Forti5 Technologies, UK and a Visiting Professor Extraordinarius at University of South Africa (UniSA). Muthu is certified assessor for IASME CE (Information Assurance for Small and Medium Enterprises Cyber Essentials Certification Assessor), CE assurance, IoT Cyber, and IoT Assurance. Muthu is also an Editorial Board member and Corporate Advisor Council Lead on AI Ethics in Healthcare at Blockchain in Healthcare Today (BHTY, USA) and Editor-in-Chief (EiC) at International Journal of Organizational and Collective Intelligence (IJOCI). Muthu has been recognized as one of the Top Blockchain Voices by LinkedIn. Muthu has 35 + years of experience in teaching & research in both academia and in industrial settings. Muthu is also the author of recent books on Blockchain Engineering in Healthcare (Springer, 2025), Engineering AI Ethics by Design (Springer, 2025), Ethics of Blockchain by Design (Elsevier, 2026), AI for Cybersecurity Vol 1 and Vol 2 (Springer 2026), and has authored and edited more than 15 books.

Declarations

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Competing Interests: None declared.

Ethics Approval: Used publicly available MIMIC-IV dataset (HIPAA de-identified). Access requires data use agreement and CITI training.

Data Availability: MIMIC-IV available at https://physionet.org/content/mimiciv/

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Figures

Figure 1

The Five Ways Framework for AI Ethics in Healthcare

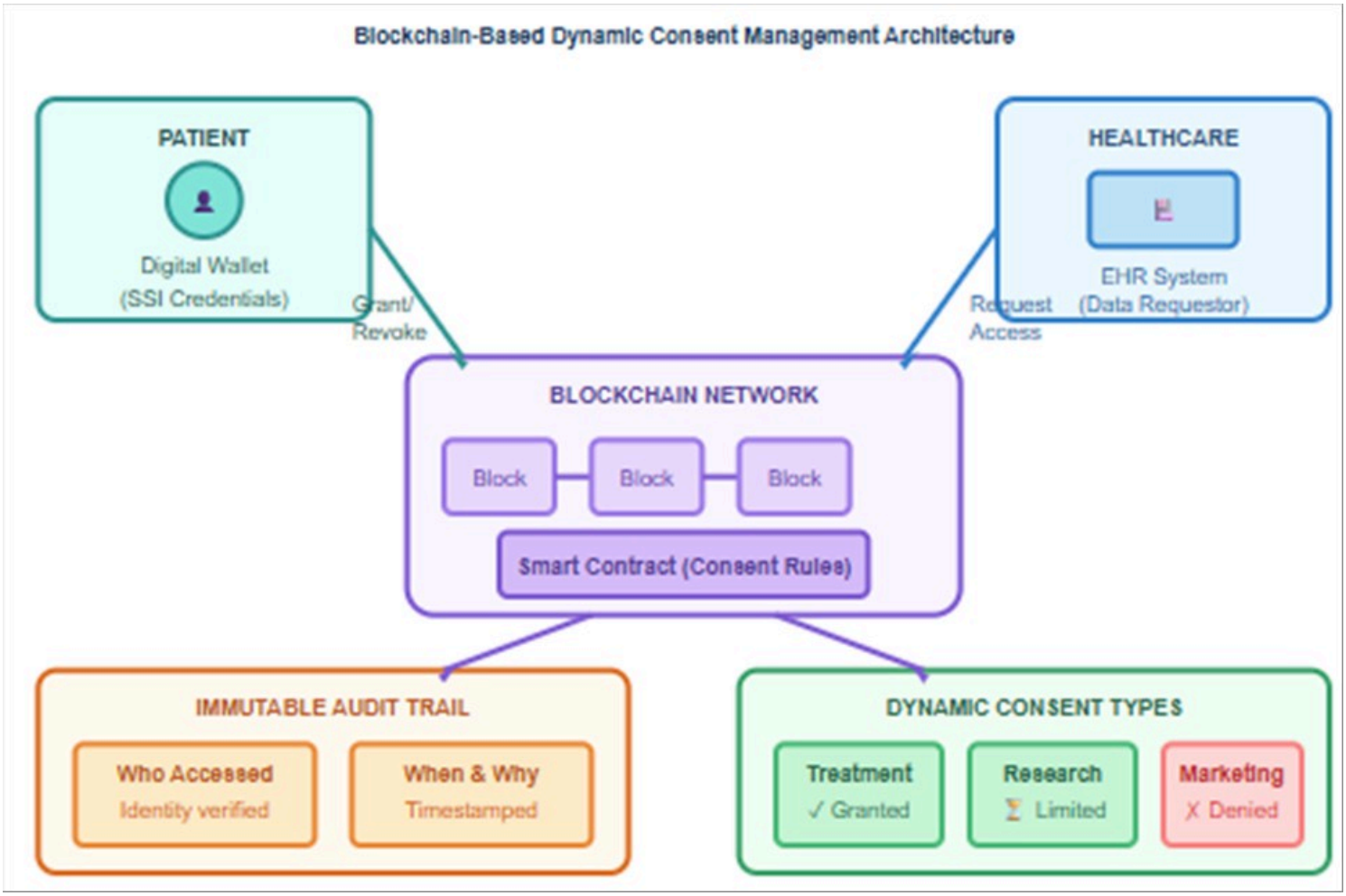

Figure 2

Blockchain-based dynamic consent management architecture

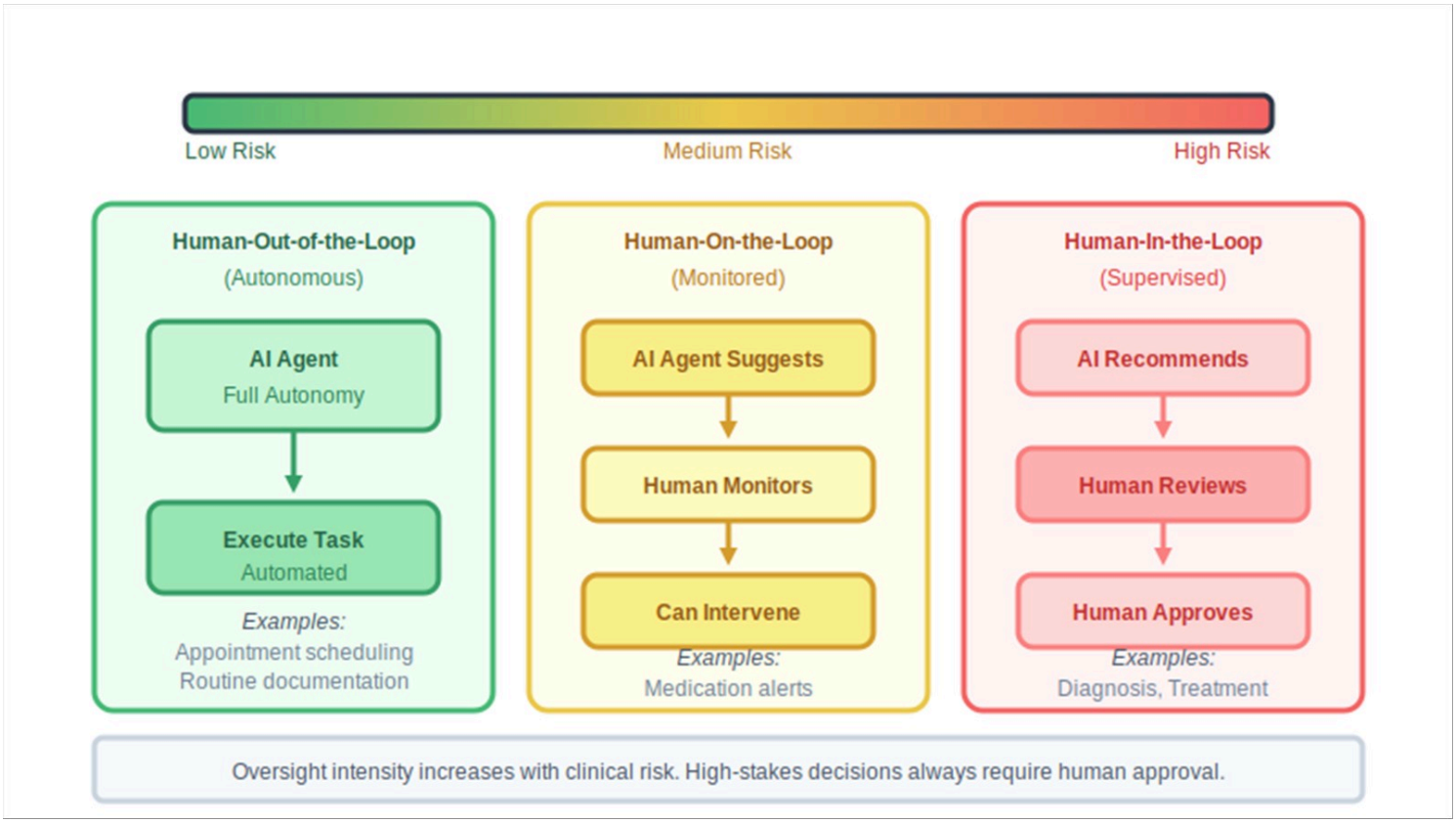

Figure 3

Human oversight architecture for autonomous healthcare AI agents.

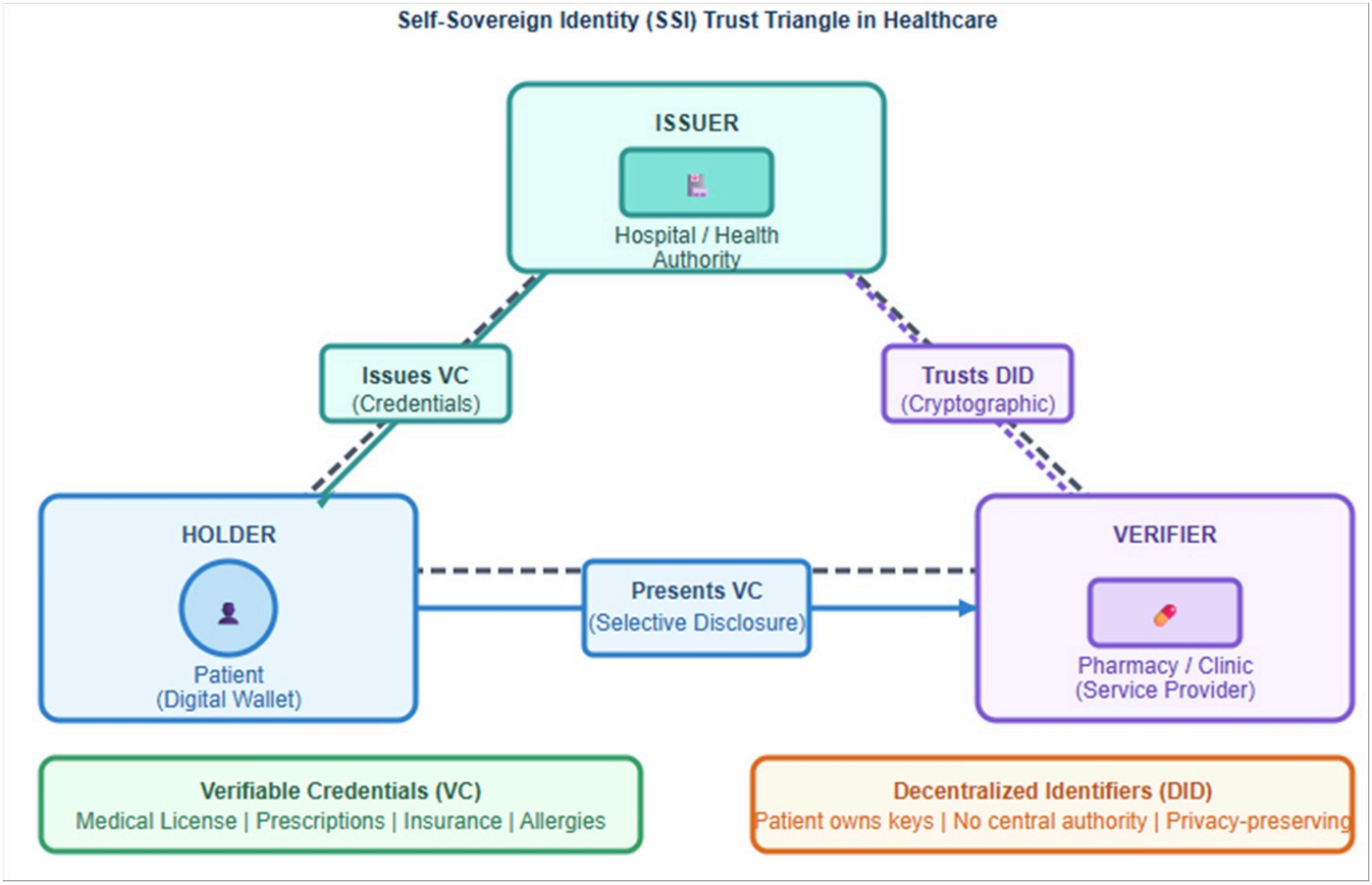

Figure 4

Self-sovereign identity (SSI) trust triangle in healthcare

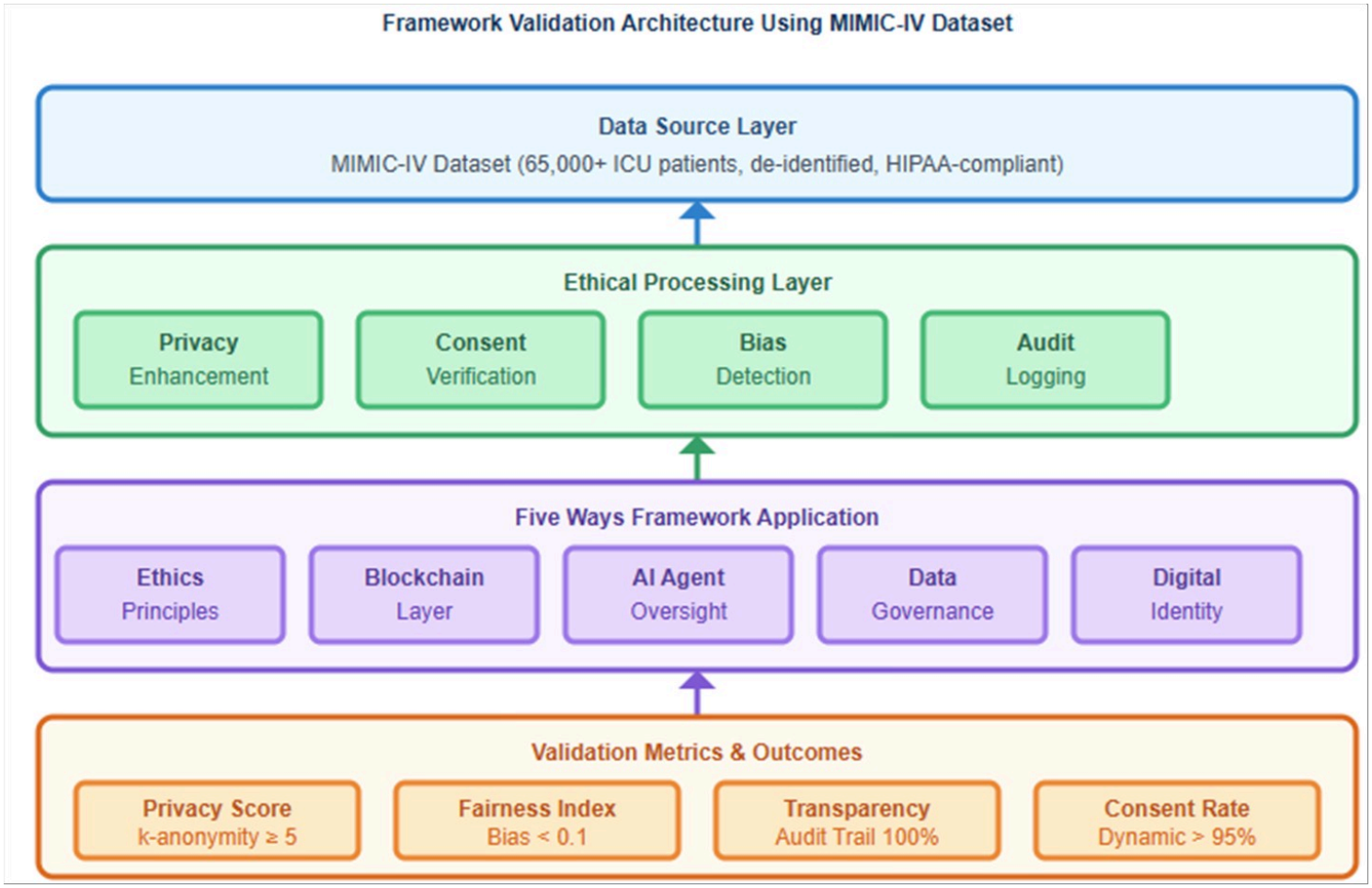

Figure 5

Framework validation architecture using MIMIC-IV dataset.